

Creation Date 03-Feb-2010

Revision Date 18-Oct-2023

Revision Number 13

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                                  |                                                  |
|----------------------------------|--------------------------------------------------|
| <b>Product Description:</b>      | <b>Trichloroethylene</b>                         |
| <b>Cat No. :</b>                 | <b>T/3101/17, T/3101/08, T/3101, T/3101/PB17</b> |
| <b>Synonyms</b>                  | Triclene; Trichloroethene; Ethylene trichloride  |
| <b>Index No</b>                  | 602-027-00-9                                     |
| <b>CAS No</b>                    | 79-01-6                                          |
| <b>EC No</b>                     | 201-167-4                                        |
| <b>Molecular Formula</b>         | C <sub>2</sub> H Cl <sub>3</sub>                 |
| <b>REACH registration number</b> | 01-2119490731-36                                 |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                       |                                                                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>                | Laboratory chemicals. Scientific research and development. REACH (1907/2006) - Annex XIV. The substance is used under strictly controlled conditions. |
| <b>Sector of use</b>                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites                                                              |
| <b>Product category</b>               | PC21 - Laboratory chemicals                                                                                                                           |
| <b>Process categories</b>             | PROC15 - Use as a laboratory reagent                                                                                                                  |
| <b>Environmental release category</b> | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)                                                           |
| <b>Uses advised against</b>           | All other uses                                                                                                                                        |

### 1.3. Details of the supplier of the safety data sheet

#### Company

#### EU entity/business name

Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

#### UK entity/business name

Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

#### Swiss distributor - Fisher Scientific AG

Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

#### E-mail address

begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

For customers in Switzerland:

Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

|                                                    |                    |
|----------------------------------------------------|--------------------|
| Skin Corrosion/Irritation                          | Category 2 (H315)  |
| Serious Eye Damage/Eye Irritation                  | Category 2 (H319)  |
| Skin Sensitization                                 | Category 1 (H317)  |
| Germ Cell Mutagenicity                             | Category 2 (H341)  |
| Carcinogenicity                                    | Category 1B (H350) |
| Specific target organ toxicity - (single exposure) | Category 3 (H336)  |

##### Environmental hazards

|                          |                   |
|--------------------------|-------------------|
| Chronic aquatic toxicity | Category 3 (H412) |
|--------------------------|-------------------|

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

Danger

#### **Hazard Statements**

H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H319 - Causes serious eye irritation  
H336 - May cause drowsiness or dizziness  
H341 - Suspected of causing genetic defects  
H350 - May cause cancer  
H412 - Harmful to aquatic life with long lasting effects

#### **Precautionary Statements**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention  
P337 + P313 - If eye irritation persists: Get medical advice/attention  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

P312 - Call a POISON CENTER or doctor if you feel unwell  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Additional EU labelling

Restricted to professional users

### 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component         | CAS No  | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                       |
|-------------------|---------|-------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trichloroethylene | 79-01-6 | EEC No. 201-167-4 | <=100    | Skin Irrit. 2 (H315)<br>Eye Irrit. 2 (H319)<br>Skin Sens. 1 (H317)<br>STOT SE 3 (H336)<br>Muta. 2 (H341)<br>Carc. 1B (H350)<br>Aquatic Chronic 3 (H412) |

REACH registration number

01-2119490731-36

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                              |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

May cause allergic skin reaction. Inhalation of high vapor concentrations may cause

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## 4.3. Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Chlorine, Phosgene, Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride gas.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### 6.2. Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from light. Do not store in aluminum containers.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1C

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

| Component         | European Union                                                                                                          | The United Kingdom                                                                                                                  | France                                                                                                                                                                                                                           | Belgium                                                                                                                               | Spain                                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trichloroethylene | TWA: 54.7 mg/m <sup>3</sup> (8h)<br>TWA: 10 ppm (8h)<br>Skin<br>STEL: 164.1 mg/m <sup>3</sup> (8h)<br>STEL: 30 ppm (8h) | STEL: 150 ppm 15 min<br>STEL: 820 mg/m <sup>3</sup> 15 min<br>TWA: 100 ppm 8 hr<br>TWA: 550 mg/m <sup>3</sup> 8 hr<br>Carc.<br>Skin | TWA / VME: 10 ppm (8 heures). restrictive limit<br>TWA / VME: 54.7 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 30 ppm. restrictive limit<br>STEL / VLCT: 164.1 mg/m <sup>3</sup> . restrictive limit<br>Peau | TWA: 10 ppm 8 uren<br>TWA: 54.7 mg/m <sup>3</sup> 8 uren<br>STEL: 25 ppm 15 minuten<br>STEL: 137 mg/m <sup>3</sup> 15 minuten<br>Huid | STEL / VLA-EC: 30 ppm (15 minutos).<br>STEL / VLA-EC: 164.1 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 10 ppm (8 horas)<br>TWA / VLA-ED: 54.7 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component         | Italy                                                                                                                                                                                                            | Germany | Portugal                                                                                                                                  | The Netherlands                                                                        | Finland                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trichloroethylene | TWA: 54.7 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>TWA: 10 ppm 8 ore.<br>Time Weighted Average<br>STEL: 164.1 mg/m <sup>3</sup> 15 minuti. Short-term<br>STEL: 30 ppm 15 minuti. Short-term<br>Pelle | Haut    | STEL: 30 ppm 15 minutos<br>STEL: 164.1 mg/m <sup>3</sup> 15 minutos<br>TWA: 10 ppm 8 horas<br>TWA: 54.7 mg/m <sup>3</sup> 8 horas<br>Pele | huid<br>STEL: 164.1 mg/m <sup>3</sup> 15 minuten<br>TWA: 54.7 mg/m <sup>3</sup> 8 uren | TWA: 10 ppm 8 tunteina<br>TWA: 54.7 mg/m <sup>3</sup> 8 tunteina<br>STEL: 164.1 mg/m <sup>3</sup> 15 minuutteina<br>STEL: 30 ppm 15 minuutteina<br>Iho |

| Component         | Austria                                                                             | Denmark                                                                                                               | Switzerland                                                                    | Poland                                                                           | Norway                                                                                                                        |
|-------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Trichloroethylene | TRK-KZGW: 2.4 ppm 15 Minuten<br>TRK-KZGW: 13.2 mg/m <sup>3</sup> 15 Minuten<br>Haut | TWA: 6 ppm 8 timer<br>TWA: 33 mg/m <sup>3</sup> 8 timer<br>STEL: 164 mg/m <sup>3</sup> 15 minutter<br>STEL: 30 ppm 15 | Haut/Peau<br>STEL: 50 ppm 15 Minuten<br>STEL: 273 mg/m <sup>3</sup> 15 Minuten | STEL: 100 mg/m <sup>3</sup> 15 minutach<br>TWA: 50 mg/m <sup>3</sup> 8 godzinach | TWA: 33 mg/m <sup>3</sup> 8 timer<br>TWA: 6 ppm 8 timer<br>STEL: 164 mg/m <sup>3</sup> 15 minutter. value from the regulation |

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

|  |                                                    |                 |                                                                     |  |                                                                  |
|--|----------------------------------------------------|-----------------|---------------------------------------------------------------------|--|------------------------------------------------------------------|
|  | TRK-TMW: 0.6 ppm<br>TRK-TMW: 3.3 mg/m <sup>3</sup> | minutter<br>Hud | TWA: 20 ppm 8<br>Stunden<br>TWA: 110 mg/m <sup>3</sup> 8<br>Stunden |  | STEL: 30 ppm 15<br>minutter. value from the<br>regulation<br>Hud |
|--|----------------------------------------------------|-----------------|---------------------------------------------------------------------|--|------------------------------------------------------------------|

| Component         | Bulgaria                                                                                                       | Croatia                                                                                                                                                                         | Ireland                                                                                                                          | Cyprus                                                                                                                                    | Czech Republic                                                                                                       |
|-------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Trichloroethylene | TWA: 54.7 mg/m <sup>3</sup><br>TWA: 10 ppm<br>STEL : 164.1 mg/m <sup>3</sup><br>STEL : 30 ppm<br>Skin notation | kože<br>TWA-GVI: 10 ppm 8<br>satima.<br>TWA-GVI: 54.7 mg/m <sup>3</sup> 8<br>satima.<br>STEL-KGVI: 30 ppm 15<br>minutama.<br>STEL-KGVI: 164.1<br>mg/m <sup>3</sup> 15 minutama. | TWA: 10 ppm 8 hr.<br>TWA: 54.7 mg/m <sup>3</sup> 8 hr.<br>STEL: 30 ppm 15 min<br>STEL: 164.1 mg/m <sup>3</sup> 15<br>min<br>Skin | Skin-potential for<br>cutaneous absorption<br>STEL: 164.1 mg/m <sup>3</sup><br>STEL: 30 ppm<br>TWA: 10 ppm<br>TWA: 54.7 mg/m <sup>3</sup> | TWA: 250 mg/m <sup>3</sup> 8<br>hodinách.<br>Potential for cutaneous<br>absorption<br>Ceiling: 750 mg/m <sup>3</sup> |

| Component         | Estonia                                                                                                                                                         | Gibraltar | Greece                                                                                                                                      | Hungary                                                                                                                                        | Iceland                                                                                                                                                  |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trichloroethylene | Nahk<br>TWA: 10 ppm 8<br>tundides.<br>TWA: 50 mg/m <sup>3</sup> 8<br>tundides.<br>STEL: 25 ppm 15<br>minutites.<br>STEL: 140 mg/m <sup>3</sup> 15<br>minutites. |           | skin - potential for<br>cutaneous absorption<br>STEL: 30 ppm<br>STEL: 164.1 mg/m <sup>3</sup><br>TWA: 10 ppm<br>TWA: 54.7 mg/m <sup>3</sup> | STEL: 164.1 mg/m <sup>3</sup> 15<br>percekben. CK<br>TWA: 54.7 mg/m <sup>3</sup> 8<br>óraban. AK<br>lehetséges borön<br>keresztüli felszívódás | STEL: 30.0 ppm<br>STEL: 164.1 mg/m <sup>3</sup><br>TWA: 10.0 ppm 8<br>klukkustundum.<br>TWA: 54.7 mg/m <sup>3</sup> 8<br>klukkustundum.<br>Skin notation |

| Component         | Latvia                                                                                                                                    | Lithuania                                                                                                                                                                                                                                                                                                                                                                                                                              | Luxembourg | Malta | Romania                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Trichloroethylene | skin - potential for<br>cutaneous exposure<br>STEL: 164.1 mg/m <sup>3</sup><br>STEL: 20 ppm<br>TWA: 54.7 mg/m <sup>3</sup><br>TWA: 10 ppm | TWA: 10 ppm IPRD<br>Trichlorethylene may<br>contain special<br>stabilizing agents at low<br>concentrations, such as<br>epichlorohydrin, which is<br>banned in some<br>countries<br>TWA: 50 mg/m <sup>3</sup> IPRD<br>Trichlorethylene may<br>contain special<br>stabilizing agents at low<br>concentrations, such as<br>epichlorohydrin, which is<br>banned in some<br>countries<br>Oda<br>STEL: 25 ppm<br>STEL: 140 mg/m <sup>3</sup> |            |       | Skin notation<br>TWA: 18.5 ppm 8 ore<br>TWA: 100 mg/m <sup>3</sup> 8 ore<br>STEL: 28 ppm 15<br>minute<br>STEL: 150 mg/m <sup>3</sup> 15<br>minute |

| Component         | Russia                                                      | Slovak Republic                                                                                                                                      | Slovenia                                                                                                                                         | Sweden                                                                                                                                                                       | Turkey |
|-------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Trichloroethylene | TWA: 10 mg/m <sup>3</sup> 2108<br>MAC: 30 mg/m <sup>3</sup> | TWA: 50 ppm 8<br>hodinách<br>TWA: 275 mg/m <sup>3</sup> 8<br>hodinách<br>STEL: 250 ppm 15<br>minútach<br>STEL: 1375 mg/m <sup>3</sup> 15<br>minútach | TWA: 54.7 mg/m <sup>3</sup> 8<br>urah<br>TWA: 10 ppm 8 urah<br>Koža<br>STEL: 164.1 mg/m <sup>3</sup> 15<br>minutah<br>STEL: 30 ppm 15<br>minutah | Binding STEL: 25 ppm<br>15 minuter<br>Binding STEL: 140<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 10 ppm 8 timmar.<br>NGV<br>TLV: 54 mg/m <sup>3</sup> 8<br>timmar. NGV<br>Hud |        |

## Biological limit values

List source(s):

| Component         | European Union | United Kingdom | France                                                                                                          | Spain                                                                                                      | Germany |
|-------------------|----------------|----------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------|
| Trichloroethylene |                |                | Free Trichloroethanol: 4<br>mg/L blood end of shift<br>at end of workweek<br>Sum of Trichloroacetic<br>acid and | Trichloroacetic acid: 15<br>mg/L urine end of<br>workweek<br>Trichloroethanol<br>(without hydrolysis): 0.5 |         |

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

|  |  |  |                                                                                                                                                |                            |  |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--|
|  |  |  | Trichloroethanol: 300 mg/g creatinine urine end of shift at end of workweek<br>Trichloroacetic acid: 100 mg/g creatinine urine end of workweek | mg/L blood end of workweek |  |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--|

| Component         | Italy | Finland                                                                                 | Denmark | Bulgaria | Romania                                                               |
|-------------------|-------|-----------------------------------------------------------------------------------------|---------|----------|-----------------------------------------------------------------------|
| Trichloroethylene |       | Trichloroacetic acid: 120 µmol/L urine after the shift at the end of a exposure period. |         |          | Trichloroacetic acid: 20 mg/L urine end of shift and end of work week |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                             | Fresh water                        | Fresh water sediment                                        | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)                                   |
|---------------------------------------|------------------------------------|-------------------------------------------------------------|--------------------|------------------------------------|------------------------------------------------------|
| Trichloroethylene<br>79-01-6 ( ≤100 ) | PNEC = 115µg/L<br>PNEC = 0.115mg/L | PNEC = 316µg/kg sediment dw<br>PNEC = 2.04mg/kg sediment dw | PNEC = 0.208mg/L   | PNEC = 2.6mg/L                     | PNEC = 155µg/kg soil dw<br>PNEC = 0.344mg/kg soil dw |

| Component                             | Marine water      | Marine water sediment         | Marine water Intermittent | Food chain            | Air |
|---------------------------------------|-------------------|-------------------------------|---------------------------|-----------------------|-----|
| Trichloroethylene<br>79-01-6 ( ≤100 ) | PNEC = 0.0115mg/L | PNEC = 0.204mg/kg sediment dw |                           | PNEC = 13.8mg/kg food |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

| Glove material           | Breakthrough time | Glove thickness        | EU standard | Glove comments                                                                 |
|--------------------------|-------------------|------------------------|-------------|--------------------------------------------------------------------------------|
| Viton (R)                | > 480 minutes     | 0.7 mm                 | EN 374      | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| PVA                      | > 360 minutes     | 0.3 mm                 |             |                                                                                |
| Nitrile rubber           | < 12 minutes      | 0.7mm                  |             |                                                                                |
| Laminated film (Barrier) | > 480 minutes     | 2.5 mil                |             |                                                                                |
| Skin and body protection |                   | Long sleeved clothing. |             |                                                                                |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

## Small scale/Laboratory use

Maintain adequate ventilation Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

## Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                                   |                                          |
|------------------------------------------------|---------------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                                            |                                          |
| <b>Appearance</b>                              | Colorless                                         |                                          |
| <b>Odor</b>                                    | Characteristic                                    |                                          |
| <b>Odor Threshold</b>                          | No data available                                 |                                          |
| <b>Melting Point/Range</b>                     | -85 °C / -121 °F                                  |                                          |
| <b>Softening Point</b>                         | No data available                                 |                                          |
| <b>Boiling Point/Range</b>                     | 87 °C / 188.6 °F                                  | Literature reference                     |
| <b>Flammability (liquid)</b>                   | No data available                                 |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable                                    | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 8,0 vol %<br><b>Upper</b> 44.8 vol % | Literature reference                     |
| <b>Flash Point</b>                             | No information available                          | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 410 °C / 770 °F                                   | DIN 51794                                |
| <b>Decomposition Temperature</b>               | > 120°C                                           |                                          |
| <b>pH</b>                                      | No information available                          |                                          |
| <b>Viscosity</b>                               | 0.55 mPa.s (25°C)                                 | Based on available literature            |
| <b>Water Solubility</b>                        | Insoluble                                         |                                          |
| <b>Solubility in other solvents</b>            | No information available                          |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                   |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                    |                                          |
| Trichloroethylene                              | 2.4                                               |                                          |
| <b>Vapor Pressure</b>                          | 77.3 mbar @ 20 °C                                 |                                          |
| <b>Density / Specific Gravity</b>              | 1.460                                             |                                          |



# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

|                          |                         |                      |
|--------------------------|-------------------------|----------------------|
| Bulk Density             | Not applicable          | Liquid               |
| Vapor Density            | 4.5 (Air = 1.0)         | Literature reference |
| Particle characteristics | Not applicable (liquid) |                      |

## 9.2. Other information

|                      |                                                      |
|----------------------|------------------------------------------------------|
| Molecular Formula    | C <sub>2</sub> H Cl <sub>3</sub>                     |
| Molecular Weight     | 131.39                                               |
| Explosive Properties | Not explosive explosive air/vapour mixtures possible |
| Oxidizing Properties | Not oxidising                                        |
| Evaporation Rate     | 0.69 (Carbon Tetrachloride = 1.0)                    |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Light sensitive.

### 10.3. Possibility of hazardous reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Exposure to light. Exposure to moist air or water.

### 10.5. Incompatible materials

Strong oxidizing agents. Strong bases. Amines. Alkali metals. Metals. . Powdered aluminum. Powdered zinc. Powdered magnesium.

### 10.6. Hazardous decomposition products

Chlorine. Phosgene. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component         | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation            |
|-------------------|---------------------------|-------------------------------|----------------------------|
| Trichloroethylene | LD50 = 4920 mg/kg ( Rat ) | LD50 = 29000 mg/kg ( Rabbit ) | LC50 = 26 mg/L ( Rat ) 4 h |

#### (b) skin corrosion/irritation;

Test method

Category 2

Test species

OECD 404

Observational endpoint

rabbit

Irritating

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory

Skin

Based on available data, the classification criteria are not met

Category 1

| Component                              | Test method             | Test species | Study result  |
|----------------------------------------|-------------------------|--------------|---------------|
| Trichloroethylene<br>79-01-6 ( <=100 ) | OECD Test Guideline 429 | mouse        | Sensitization |

May cause sensitization by skin contact

(e) germ cell mutagenicity;

Category 2

Mutagenic effects have occurred in humans

(f) carcinogenicity;

Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component         | EU           | UK | Germany | IARC    |
|-------------------|--------------|----|---------|---------|
| Trichloroethylene | Carc Cat. 1B |    | Cat. 1  | Group 1 |

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Category 3

Results / Target organs

Central nervous system (CNS).

(i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## 11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Do not empty into drains. The product contains following substances which are hazardous for the environment. Contains a substance which is: Harmful to aquatic organisms. Toxic to aquatic organisms.

| Component         | Freshwater Fish                                          | Water Flea                            | Freshwater Algae                                           |
|-------------------|----------------------------------------------------------|---------------------------------------|------------------------------------------------------------|
| Trichloroethylene | LC50: 31.4 - 71.8 mg/L, 96h<br>flow-through (Pimephales) | EC50: = 2.2 mg/L, 48h (Daphnia magna) | EC50: = 175 mg/L, 96h<br>(Pseudokirchneriella subcapitata) |

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

|  |                                                                      |  |                                                    |
|--|----------------------------------------------------------------------|--|----------------------------------------------------|
|  | promelas)<br>LC50: 39 - 54 mg/L, 96h static<br>(Lepomis macrochirus) |  | EC50: = 450 mg/L, 96h<br>(Desmodesmus subspicatus) |
|--|----------------------------------------------------------------------|--|----------------------------------------------------|

| Component         | Microtox                                                                                                                                           | M-Factor |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Trichloroethylene | EC50 = 0.81 mg/L 24 h<br>EC50 = 115 mg/L 10 min<br>EC50 = 190 mg/L 15 min<br>EC50 = 235 mg/L 24 h<br>EC50 = 410 mg/L 24 h<br>EC50 = 975 mg/L 5 min |          |

## 12.2. Persistence and degradability

**Persistence** Persistence is unlikely, based on information available.  
**Degradability** See values below.

| Component                              | Degradability         |
|----------------------------------------|-----------------------|
| Trichloroethylene<br>79-01-6 ( <=100 ) | 2.4 % (14d) OECD 301C |

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Trichloroethylene | 2.4     | 90(Fish)                      |

## 12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

## 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

## 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**European Waste Catalogue (EWC)** According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information** Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

chemical enter the environment.

## Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

**14.1. UN number** UN1710  
**14.2. UN proper shipping name** TRICHLOROETHYLENE  
**14.3. Transport hazard class(es)** 6.1  
**14.4. Packing group** III

### ADR

**14.1. UN number** UN1710  
**14.2. UN proper shipping name** TRICHLOROETHYLENE  
**14.3. Transport hazard class(es)** 6.1  
**14.4. Packing group** III

### IATA

**14.1. UN number** UN1710  
**14.2. UN proper shipping name** TRICHLOROETHYLENE  
**14.3. Transport hazard class(es)** 6.1  
**14.4. Packing group** III

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component         | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL | ENCS | ISHL |
|-------------------|---------|-----------|--------|-----|-------|------|------|------|------|
| Trichloroethylene | 79-01-6 | 201-167-4 | -      | -   | X     | X    | X    | X    | X    |

| Component         | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-------------------|---------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Trichloroethylene | 79-01-6 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

| Component         | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization                                                           | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Trichloroethylene | 79-01-6 | Carcinogenic Category 1B, Article 57<br>Application date: October 21, 2014<br>Sunset date: April 21, 2016<br>Exemption - None | Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - 201-167-4 - Carcinogenic, Article 57a                                           |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

## REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Seveso III Directive (2012/18/EC)

| Component         | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Trichloroethylene | 79-01-6 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WKG Classification

See table for values

| Component         | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                                           |
|-------------------|---------------------------------------|-----------------------------------------------------------------------------------|
| Trichloroethylene | WGK3                                  | Krebserzeugende Stoffe - Class III : 1 mg/m <sup>3</sup><br>(Massenkonzentration) |

| Component         | France - INRS (Tables of occupational diseases)                    |
|-------------------|--------------------------------------------------------------------|
| Trichloroethylene | Tableaux des maladies professionnelles (TMP) - RG 3, RG 12, RG 101 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component | Switzerland - Ordinance on the | Switzerland - Ordinance on | Switzerland - Ordinance of the |
|-----------|--------------------------------|----------------------------|--------------------------------|
|-----------|--------------------------------|----------------------------|--------------------------------|

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

|                                       | Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Incentive Taxes on Volatile Organic Compounds (OVOC) | Rotterdam Convention on the Prior Informed Consent Procedure |
|---------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------|
| Trichloroethylene<br>79-01-6 ( ≤100 ) | Prohibited and Restricted Substances                                            | Group I                                              |                                                              |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted by the manufacturer/importer

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H319 - Causes serious eye irritation  
H336 - May cause drowsiness or dizziness  
H341 - Suspected of causing genetic defects  
H350 - May cause cancer  
H412 - Harmful to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Chemical incident response training.

**Creation Date**

03-Feb-2010

**Revision Date**

18-Oct-2023

**Revision Summary**

Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No**

# SAFETY DATA SHEET

Trichloroethylene

Revision Date 18-Oct-2023

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1907/2006 .

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**